

POOJA AWASARE

Civil & Environmental Engineer · Circular Construction & Digital Decision-Support

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Location: Pune, India  
Languages: English (professional); learning Dutch

Professional Summary

Civil and environmental engineer focused on the **digital transformation of circular construction**. My work sits where sustainability, digital tools, and construction practice meet: using **BIM, GIS, digital twins, and material-passport data** to make reuse and **reverse-logistics** decisions clearer when availability, quality, location, timing, transport, cost, and CO<sub>2</sub> are uncertain. My Master's thesis on Agile Project Management in the construction industry showed me that transforming construction depends as much on **transparent information flow and stakeholder coordination** as on technical innovation, and led me to a bigger question: what happens to materials at end-of-life, and how do we get reliable information about them to everyone who needs it? That question moved my focus from improving project delivery to **designing data-driven decision-support for circular construction**. With hands-on experience in digital twins, data integration, requirements engineering, and Agile delivery across manufacturers and construction clients, I want to help make circular construction **practical, data-driven, and widely adopted**, which is exactly the aim of the CIRCOLOGIC EngD.

Education

|           |                                                                                                                                                                                                                    |        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 2018–2022 | <b>HTW Dresden (Hochschule für Technik und Wirtschaft), Germany</b><br>Master of Engineering (M.Eng), Environmental Engineering<br><i>Thesis: Agile Project Management for Sustainable Construction Innovation</i> | 2.8    |
| 2014–2017 | <b>Savitribai Phule Pune University, India</b><br>Bachelor of Engineering (B.E), Civil Engineering<br><i>Structural design, project planning &amp; environmental studies</i>                                       | 65.26% |
| 2010–2014 | <b>Maharashtra State Board of Technical Education (MSBTE), India</b><br>Diploma, Civil Engineering                                                                                                                 | 68.11% |
| 2010      | <b>Maharashtra State Board (MSBSHSE), India</b><br>10th Standard (Secondary School Certificate)                                                                                                                    | 82.55% |

Professional Experience and Training

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun'24 – Present | <b>Career Break</b> · Belgium & India <ul style="list-style-type: none"><li>Planned career break for parenting newborns, spanning relocation from Belgium to India. Alongside family responsibilities, maintained professional currency through independent study in <b>circular construction and reverse logistics</b> and <b>GIS / geospatial and machine-learning</b> (Google Earth Engine), preparing for design-science doctoral research.</li></ul>                                                    |
| Aug'23 – Jun'24  | <b>Civil &amp; Environmental Project Consultant, iosttring, Belgium</b> <ul style="list-style-type: none"><li>Delivered civil projects end to end with a focus on time, cost and value, using project data to guide <b>material and reuse choices that reduce waste</b>, treating circularity as part of delivery rather than a side track.</li><li>Defined problem statements, use cases and solution architecture, and linked and analysed project data to support material and reuse decisions.</li></ul> |
| Nov'22 – Aug'23  | <b>Project Engineer, Hive CPQ, Ghent, Belgium</b> <ul style="list-style-type: none"><li>Managed large datasets and <b>gathered requirements</b> from manufacturers (Pellenc, Mecatherm, Soproda) and customers, integrating manufacturing, construction and supply-chain data into a data mart.</li><li>Built configurators that <b>turned complex rules into intuitive digital tools</b>, translating stakeholder needs into practical solutions for technical and non-technical users.</li></ul>           |

Jun'22 – Nov'22

**Project Engineer, Aug.e, Brussels, Belgium**

- Contributed to energy-transition and smart-building projects using a **digital-twin stack** (Microsoft Azure, Navisworks Manage, Neanex, Revit), organising and integrating data from physical assets into digital decision-support.
- Worked with IoT, AI and metadata layers linking building and energy data to decisions, close to the data-integration mindset behind **BIM, product passports and logistics parameters**; tracked delivery in Azure DevOps.

Aug'21 – Sep'21

**Trainee (Master's Thesis), HTW Dresden, Germany**

- Researched **Agile project planning in an IT-sustainability setting** aligned to UN Global Compact / SDG themes; gathered qualitative and quantitative requirements on environmental impacts (climate, waste, water, energy) and built product backlogs in Jira.

Oct'19

**Trainee, Fraunhofer Institute, Dresden, Germany**

- Planned a **green business model** for biogas-to-biowax demand and adopted EU sustainability policy, assessing green-business financials and transformation tools.

## Technical Skills

|                            |                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Digital &amp; BIM</b>   | Revit, Navisworks Manage, Neanex, AutoCAD; digital twins, product passports                                  |
| <b>Geospatial / GIS</b>    | Google Earth Engine, GIS-style mapping, change detection, machine learning                                   |
| <b>Data &amp; analysis</b> | Data modelling & integration, Tableau, MongoDB, Postman, Lucidchart, Excel                                   |
| <b>Project &amp; Agile</b> | Jira, Confluence, Asana, Miro, Azure DevOps, Primavera P6; Scrum, Kanban (PSK I)                             |
| <b>Domain</b>              | Circular construction, reverse logistics, LCA / CO <sub>2</sub> awareness, EU sustainability policy, UN SDGs |
| <b>Research methods</b>    | Design-science research, qualitative case studies, requirements engineering                                  |

## Certifications

- Professional Scrum with Kanban (PSK I) · Scrum.org
- Google Earth Engine for Machine Learning & Change Detection
- Construction Management: Planning and Scheduling
- Primavera P6
- Excel Essential Training (Office 365 / Microsoft 365)

## Publications

"Design and Analysis of Building by AKSES and Manual Method."

## Areas of Interest

- Circular construction: reuse, refurbishment and end-of-life material recovery
- Reverse logistics and construction supply chains
- Digital decision-support: BIM, GIS, digital twins and material passports
- Design-science research for sustainability and CO<sub>2</sub> reduction

## References

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|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| • <b>Frederik Taleman</b><br>CEO, Hive-CPQ                                           | • <b>Prof. Dr. rer. nat. Martin Oczipka</b><br>HTW Dresden, Germany |
| • <b>Micheal Dhont</b><br>Consultant, McKinsey & Company; former Product Head, Aug.e | • <b>Khushal Kanade</b><br>Assistant Professor, MIT, Pune, India    |